SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I __N.Umasri(192424240)__ (Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:N.Umasri

1. Smart Campus Subnet Allocation

Given Network: 192.168.100.0/24

(1) Network Address and Broadcast Address

Department	**Subnet Mask**	**Network Address**	**Broadcast Address**
School of Computing	/25	192.168.100.0	192.168.100.127
School of Electronics	/26	192.168.100.128	192.168.100.191
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223

(2) Valid Host IP Range

Department	**Host Range**
School of Computing	192.168.100.1 – 192.168.100.126
School of Electronics	192.168.100.129 – 192.168.100.190
School of Mechanical Engineering	192.168.100.193 – 192.168.100.222

(3) Number of Usable Hosts

Department	**Usable Hosts**
School of Computing	126
School of Electronics	62
School of Mechanical Engineering	30

(4) Remaining Unused IP Range

Unused addresses:

192.168.100.224 – 192.168.100.255

(32 IP addresses remain unused.)

2. Multinational Software Company

Given Network: 10.10.0.0/16

(1) Network Address and Broadcast Address

Department	Subnet	Network Address	Broadcast Address
Software Development	/24	10.10.0.0	10.10.0.255
Quality Assurance	/25	10.10.1.0	10.10.1.127
Human Resources	/26	10.10.1.128	10.10.1.191
Executive Management	/27	10.10.1.192	10.10.1.223

(2) Valid Host Range

Department	Host Range
Software Development	10.10.0.1 – 10.10.0.254
Quality Assurance	10.10.1.1 – 10.10.1.126
Human Resources	10.10.1.129 – 10.10.1.190
Executive Management	10.10.1.193 – 10.10.1.222

(3) Number of Usable Hosts

Department	Usable Hosts
Software Development	254
Quality Assurance	126
Human Resources	62
Executive Management	30

(4) Remaining Unused IP Range

Unused addresses after allocations:

10.10.1.224 – 10.10.255.255

3. Smart Hospital IPv6

Given Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

(1) Compressed IPv6 Address

2001:db8:abcd::/64

(2) Expanded Address

2001:0db8:abcd:0000:0000:0000:0000:0000

(3) Network Prefix

2001:db8:abcd:0::/64

(4) Total Host Addresses Supported

Host bits = 128 − 64 = **64**

Total hosts

2^{64} = 18,446,744,073,709,551,616

(5) Total Possible Host Addresses

18,446,744,073,709,551,616 (2^{64})

4. Routing Table

Routing Table

- 192.168.1.0/24

- 192.168.2.0/24
- 192.168.3.0/24

Destination IP

192.168.2.45

(1) Destination Subnet

192.168.2.0/24

(2) Matching Routing Entry

192.168.2.0/24

(3) Next-Hop Network/Interface

Forward through the interface connected to the **192.168.2.0/24** network.

(4) Default Route

If no route matches:

0.0.0.0/0 (Default Route)

The router forwards the packet to the default gateway.

5. Company LAN Configuration

Administration LAN

192.168.10.0/26

Finance LAN

192.168.10.64/26

(1) Network and Broadcast Address

LAN	Network Address	Broadcast Address
Administration	192.168.10.0	192.168.10.63
Finance	192.168.10.64	192.168.10.127

(2) Valid Host Range

LAN	Host Range
Administration	192.168.10.1 – 192.168.10.62
Finance	192.168.10.65 – 192.168.10.126

(3) Suggested IP Addresses

Administration LAN

- PC1 → **192.168.10.2**
- PC2 → **192.168.10.3**
- PC3 → **192.168.10.4**

Finance LAN

- PC1 → **192.168.10.66**
- PC2 → **192.168.10.67**
- PC3 → **192.168.10.68**

(4) Default Gateway

LAN	Default Gateway
Administration	**192.168.10.1**
Finance	**192.168.10.65**